

CSE 309 Theory of Automata, Spring 2026
Exercise 1

1. **Preliminaries** Attempt the Chapter 0 Exercises in the class text book Introduction to the Theory of Computation by Michael Sipser, third edition (Pages 25–27).

2. Decide whether the following proof is correct and identify the proof's flaw.

Proposition: All students love homeworks equally.

Proof:

- **Base Case.** For $n = 1$, a single student clearly loves homework exactly as much as him- or herself, so the base holds.
- **Inductive Hypothesis.** Assume any k students love homework equally.
- **Inductive Step.** Consider $k + 1$ students. By the inductive hypothesis, the first k students all love homework equally. By the inductive hypothesis, the last k students all love homework equally. Therefore, all $k + 1$ students love homework equally, and the proposition then holds by principle of induction.

3. **Operations on languages**

- (a) Give examples of strings in the language $(L_1L_2)^+L_3L_4^*L_5$ where $L_1 = \{\text{he, she, my TA}\}$, $L_2 = \{\text{said, told, whispered}\}$, $L_3 = \{\text{that CSE 309 is}\}$, $L_4 = \{\text{so}\}$, and $L_5 = \{\text{much fun}\}$.
- (b) Give examples of strings in the language $L_1^*(\Sigma^*\{n\}\Sigma^*)$ where $L_1 = \{\text{I, do not, like, 309}\}$. Here $\Sigma = \{\text{I, a, b, } \dots, \text{z, 3, 0, 9}\}$.

4. **Language concatenation**

- (a) Let $L = \{\lambda, a\}$. Characterize L^{20} .
- (b) Let $L = \Sigma^* \cdot \{aa, bb\} \cdot \Sigma^*$. Characterize $\overline{L}$.
- (c) Compute L_1L_3 and L_2L_3 where $L_1 = \{\}$, $L_2 = \{\lambda\}$, and $L_3 = \{b\}$.

5. **Which language?** Let $\text{PREFIX}(L) = \{u \mid u \text{ is a prefix of } w \text{ where } w \in L\}$ be the set of all prefixes of strings in L .

- (a) Let $L = \Sigma^* \cdot \{a\}$. Characterize $\text{PREFIX}(L)$.
- (b) Let $L = \{a^n b^n \mid n \geq 0\}$. Characterize $\text{PREFIX}(L)$.
- (c) Let $L = \{a^m b^n \mid 0 \leq m \leq n\}$. Characterize $\text{PREFIX}(L)$.